```
$ python -m wad.per_detection --features yolo   (training the per-detection model)
train features (yolo, 68 columns): 3338916 detections, 81 s
model: 107 iterations, 171 s -> C:\Users\nicho\Downloads\Weather Adaptive Autonomous Driving\wad-
    harness\weights\per_detection_yolo.joblib
BDD100K: graded, 223 s
DAWN: graded, 225 s
fold 0: 104 iterations, 88 s
fold 1: 103 iterations, 86 s
fold 2: 97 iterations, 82 s
fold 3: 97 iterations, 82 s
fold 4: 102 iterations, 84 s

$ python -m wad.dial_ablation   (ablations: refits on BDD100K train, graded on BDD100K val / DAWN)
train: 3,338,916 detections on 21,831 images, 223 s
model 2 (adopted, 68 features) [100%, seed 0]: 107 iterations, 210 s fit; BDD100K lam 10 cost 0.9103 (0.9062-0.9146),
    DAWN 0.9332; ECE 0.11 / 0.98 pt; total 525 s
without class probabilities (-10) [100%, seed 0]: 113 iterations, 172 s fit; BDD100K lam 10 cost 0.9139
    (0.9097-0.9182), DAWN 0.9546; ECE 0.14 / 1.16 pt; total 706 s
context only (model 1, 20) [100%, seed 0]: 81 iterations, 61 s fit; BDD100K lam 10 cost 0.9366 (0.9329-0.9405), DAWN
    0.9377; ECE 0.12 / 0.97 pt; total 1199 s
score and class only (2) [100%, seed 0]: 77 iterations, 32 s fit; BDD100K lam 10 cost 0.9946 (0.9929-0.9964), DAWN
    0.9877; ECE 0.14 / 1.00 pt; total 1355 s
model 2, training subset [10%, seed 0]: 84 iterations, 15 s fit; BDD100K lam 10 cost 0.9340 (0.9299-0.9383), DAWN
    0.9570; ECE 0.28 / 1.07 pt; total 1376 s
done, 1852 s

$ python -m wad.crc_lambda   (pre-registered: CRC over lambda vs the per-weather baseline, 200 splits)
BDD100K: evaluated, 776 s
DAWN: evaluated, 813 s
sanity: the baseline arm reproduces run 2's 200-split per-bin summary
C1 validity: ours mean held-out miss <= alpha + 0.01 wherever the baseline holds (alpha 0.2) -> 11 conditions where
    the baseline holds; ours max mean miss 0.1984 -> met
C2 price: median per-split FPPI ratio ours/baseline < 1 in >= 9 of 11 conditions -> 11 of 11 -> met
C3 pooled price: 97.5th percentile of the pooled per-split ratio < 1 -> median 0.610, 95% 0.589-0.633 -> met
done, 831 s

$ python -m wad.detector_ablation fit   (ablation d: the pipeline refitted for four other detectors)
v8x_640 train: 1384171 detections, 202367 real, 442 s
v8x_640 model 2: 108 iterations, 568 s
v8x_640: done, 1032 s
v8s_640 BDD100K: matching identical to run 2's cache (check passed)
v8s_640 train: 1887887 detections, 185916 real, 467 s
v8s_640 model 2: 105 iterations, 626 s
v8s_640: done, 1100 s
v8s_1280 train: 3401999 detections, 219682 real, 392 s
v8s_1280 model 2: 97 iterations, 587 s
v8s_1280: done, 1020 s
rtdetr_640 train: 4497938 detections, 226756 real, 632 s
rtdetr_640 model 1: 72 iterations, 780 s
rtdetr_640: done, 1524 s

$ python -m wad.full_train scan / gpu / fit   (the model refitted on all 69,998 BDD100K train frames)
scan: 70000 frames, 69998 kept (exact duplicates dropped), 861718 objects, 2908 s
detection done, 8626 s
per-box outputs done, 17467 s
features: {'frames': 69998, 'objects': 861718, 'detections': 10313532, 'real': 740649}, 440 s
model 1: 101 iterations, 643 s
model 2: 142 iterations, 1346 s
BDD100K: graded and CRC, 1570 s
DAWN: graded and CRC, 1588 s
- pooled false-alarm ratio ours / baseline: 0.587 (0.567-0.615); adopted model (subset): 0.610 (0.589-0.633)
done, 1589 s
```